SynDrugMD: An Annotation-Based Framework for Synthetic Lethality and Pathway Analysis of Drug Combination Synergy

***Abstract*—SynDrugMD is an annotation-based framework relating drug combination synergy to synthetic-lethality (SL) and pathway information. The reported collection contains 555,002 records across 3,861 drugs and 229 cell lines. Raw measurements were reprocessed using a common protocol; collected multi-omics resources were not computationally integrated. The reported selected SL subset contains 14,041 annotation-positive records among 22,881 synergistic records (61.37%). Pathway analyses use 216,030 target-annotated records. Overlap Pathway Count (OPC) measures distinct common pathways, while Regulatory Pathway Count (RPC) counts direct pathway relations. We report group counts, mean differences, cluster-aware intervals and multiplicity-adjusted comparisons, retaining 17 original tissue/lineage labels. Raw OPC/RPC differences attenuated after source, target/pathway-size and degree standardization; intervals for the fully standardized normalized metrics included zero. Threshold profiles report both conditional class proportions and retained counts. These results describe size-sensitive annotation associations and do not establish mechanisms, independently validated prediction, or clinical efficacy.**

***Index Terms*—Drug combination synergy, synthetic-lethality annotation, pathway relationships**

I. Introduction

Combination therapy can improve treatment outcomes [1], [2], but clinical benefit, cell-assay synergy and molecular mechanism are distinct questions [3]–[6]. DrugComb [7], DrugCombDB [8] and SYNERGxDB [9] organize screening and pharmacogenomic data for complementary analyses.

SynDrugMD examines three annotation summaries: cross-drug target-pair matches to an SL reference, distinct common pathways measured by OPC, and direct pathway relations measured by RPC. The contribution is descriptive characterization of annotations associated with recorded synergy labels. Target-set size, annotation opportunity, repeated observations and cohort selection can affect these associations. We therefore report uncertainty, multiplicity and scale sensitivity alongside group differences, without claiming a predictive model or a causal drug mechanism.

II. Methods

A. Data Sources and Annotation Curation

Study data were assembled from COVID-19_Map [10], the DrugCell download resource [11], and DrugCombDB [8], together with the associated screening deposits [12]. Records describe drug combinations, cell lines, tissues, and experimental responses. Drug identities were harmonized using DrugBank identifiers, and target genes using NCBI Gene IDs. Drug-target annotations were obtained primarily from Jaaks et al. [13], with missing target information supplemented from DrugBank [14].

Integrated records were deduplicated by cell line and ordered anchor–library drug names. When labels conflicted, the first record was retained, with Jaaks et al. (2022) prioritized in the input order [13]. Target-related curation excluded combinations lacking drug-target annotations and combinations for which none of the listed targets could be resolved in PubChem [15]. PubChem lookup concerned target-annotation resolvability, not verification of SL interactions. These record-level curation steps are distinct from the within-study replicate classification described below.

The available cleaned parent table contained 216,030 records: 36,688 labeled synergistic and 179,342 labeled non-synergistic. The reported SL analysis included a further-selected subset of 22,881 synergistic records and the 179,342 non-synergistic background records. Both subsets originate from the same integrated data collection; comparable application of the subsequent selection steps to the two groups has not been fully established.

Original tissue/lineage labels from DrugComb-derived records and Jaaks et al. were retained. Bowel and Large Intestine were not merged because their experimental settings differ. All 17 supplied labels, including Unknown, are reported without imposing a common tissue ontology.

B. Uniform Measurement Quality Control and Synergy Classification

Raw measurements from all sources were reprocessed using the quality-control and classification protocol of Jaaks et al. [13]. Plate-wise viability was calculated as $v = (F - B)/(NC - B)$, where F, B, and NC denote treated-cell, blank, and negative-control fluorescence. Anchor viability was averaged across five wells. Library and combination responses were fitted with two-parameter sigmoids; combination fits reused library-derived cell-line parameters and were capped at anchor viability. Replicates were fitted within a nested model, and curves with RMSE > 0.2 were excluded. Bliss responses were computed from anchor and library viabilities. Bliss and combination IC_{50} values used the half-anchor-viability inflection point; $\mathrm{E}_{\max}$ was evaluated at the highest tested library concentration, $C_{\max}$. Synergy required combination $\mathrm{IC}_{50} < 2C_{\max}$ and either $\Delta\mathrm{IC}_{50} \geq 3$ ($\log_2$ potency shift; eightfold) or $\Delta E_{\max} \geq 0.2$ (20% viability shift), with shifts

defined as Bliss minus combination responses. At least half the replicates had to qualify at a given anchor concentration; synergy at either anchor concentration defined a synergistic combination–cell-line pair. Records lacking measurements required to verify quality or classify synergy were excluded.

C. Multi-Omics Reference Resources

Multi-omics resources were collected and manually checked as references, without computational integration into SL mapping or OPC/RPC calculations.

D. Synthetic-Lethality Annotation Mapping

Synthetic lethality describes a genetic interaction in which combined perturbation of two genes compromises cell viability while either perturbation alone is tolerated under the relevant conditions [16]–[18]. In a tumor with an existing alteration in one gene, inhibiting a synthetic-lethal partner may be therapeutically useful; this differs from assuming that any two drugs targeting an annotated pair reproduce that interaction.

Human synthetic-lethality annotations were obtained from the SynLethDB download portal [19]. The supplied `Human_SL.csv` contains 35,943 unique undirected human gene pairs, with gene identifiers, evidence sources, cell-context annotations, and source-based scores. The reference includes screening, text-mining, and computational-prediction evidence. Drug targets in the reported synergistic cohort were matched to these gene pairs in either orientation. A record was annotation-positive if at least one cross-drug target pair occurred in the reference. No reference-score cutoff or cell-context matching was applied. For the additional cleaned-cohort annotations, target lists were normalized to unique positive integer gene identifiers; records with missing or invalid target lists were not counted as annotation-negative. This procedure identifies candidate genetic relationships; it does not measure simultaneous target inhibition or demonstrate a synthetic-lethal mechanism in the tested cell line.

E. Pathway Metrics

Gene-to-pathway memberships use the September 2023 CTD snapshot [20]. Let $\mathcal{T}_A$ be drug A's target set and $\mathcal{P}(i)$ the pathways of target i. Define the distinct pathway set $\mathcal{U}_A = \bigcup_{i\in\mathcal{T}_A} \mathcal{P}(i)$, and analogously $\mathcal{U}_B$. The implemented overlap count is

$$\mathrm{OPC}(A,B) = |\mathcal{U}_A \cap \mathcal{U}_B|.$$

Each common pathway contributes once, irrespective of the number of targets mapping to it.

Direct pathway relations use the supplied 2023 Reactome snapshot [21]. The pathway hierarchy relation file was symmetrized and adjacency lists deduplicated. With $\mathcal{E}$ the resulting unordered relation set,

$$\mathrm{RPC}(A,B) = \sum_{p\in\mathcal{U}_A}\sum_{q\in\mathcal{U}_B} \mathbb{I}(\{p,q\}\in\mathcal{E}).$$

Pathway identifiers were aligned across files. Both cross-set orientations contribute when present; no transitive closure or molecular-node intersection is computed. RPC is a descriptive relation count, not measured regulation or a higher-order hypergraph effect.

F. Threshold Proportions and Support

For score $M \in \{\mathrm{OPC}, \mathrm{RPC}\}$, let $\mathcal{C}_g$ denote records with label $g \in \{\mathrm{syn}, \mathrm{non}\}$. We use a strict threshold:

$$N_g^M(\tau) = \sum_{r\in\mathcal{C}_g} \mathbb{I}(M(r) > \tau),$$

$$P_{\mathrm{syn}}^M(\tau) = \frac{N_{\mathrm{syn}}^M(\tau)}{N_{\mathrm{syn}}^M(\tau) + N_{\mathrm{non}}^M(\tau)}.$$

The denominator is all retained records from both groups. An empty denominator is undefined. Figures show thresholds retaining at least 100 records; complete counts, including the sparse tail, accompany the analysis. The reference is the unfiltered cohort proportion, not a 0.5 chance baseline.

G. Statistical Comparisons

Pathway comparisons use 36,688 synergistic and 179,342 non-synergistic records. We report mean differences, record-level Hedges g, and two-sided inference using cell-cluster CR1 sandwich standard errors with t_{G-1} intervals (G cell clusters). Case-normalized cell names give 224 clusters. Welch tests and separate clustering by source plus unordered DrugBank-ID pair provide sensitivity analyses. At least two records per group and two cell clusters are required for primary inference; otherwise results are marked not estimable (NE).

Benjamini–Hochberg correction is applied separately by inference method and metric family. Each family contains two metrics across the overall cohort and all 17 labels (36 planned tests); 32 primary comparisons are estimable. Raw and normalized families are separate. All group sizes, effects, intervals, p-values and adjusted q-values are supplied in the accompanying statistical tables. Averaging resampled p-values is not used. One-way clustering does not remove all crossed cell/drug dependence; inference remains exploratory.

H. Scale Sensitivity

We additionally use pathway Jaccard overlap $J = |\mathcal{U}_A \cap \mathcal{U}_B|/|\mathcal{U}_A \cup \mathcal{U}_B|$ and relation density $D = \mathrm{RPC}/(|\mathcal{U}_A||\mathcal{U}_B|)$, requiring both drugs to have mapped pathways. This retains 195,385 records (35,283 synergistic; 160,102 non-synergistic).

Standardization sequentially conditions on source, smaller/larger target counts (1, 2–5, 6+), and log-base-4 bins of pathway-count and degree-sum products. Degree is measured in the relation graph induced by CTD pathways. Common-support strata require at least five source-specific drug-pair clusters per group and receive pooled-record weights. Within-source pair-cluster bootstrap intervals use 500 draws (seed 20261019); draws losing a required group in a stratum are omitted and counted. These coarse adjustments are sensitivity checks, not a degree-preserving network null.

III. Results

A. Synthetic-Lethality Annotation Coverage

The SL analysis included 22,881 synergistic and 179,342 non-synergistic records originating from the same integrated data collection. At least one annotated cross-drug target pair was found in 14,041 synergistic records (61.37%; Fig. 1) and 54,126 non-synergistic records (30.18%). Because the comparability of subsequent group selection has not been fully confirmed, these percentages are presented as a descriptive comparison that may be affected by selection bias. They do not by themselves establish SL enrichment independent of selection or an SL mechanism of drug action.

The outer ring of Fig. 1 groups records by the reported number of matching target pairs. The mapping provides candidate genetic relationships for further study. It does not demonstrate target engagement, tumor-selective killing, or preservation of healthy tissue.

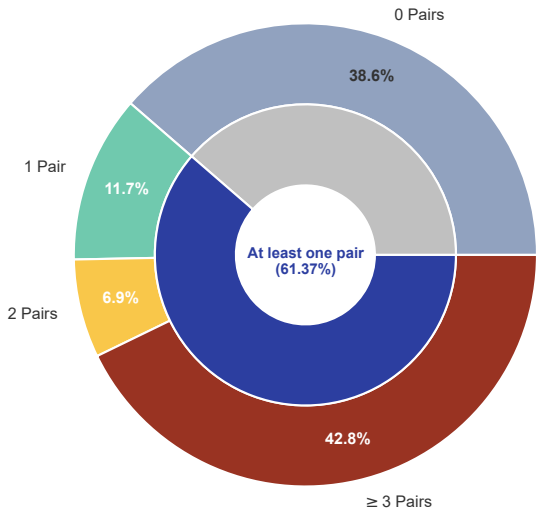


Fig. 1. Synthetic-lethality annotation coverage in the reported synergistic cohort. Of 22,881 labeled records, 14,041 (61.37%) have at least one annotated cross-drug target-pair match. The outer ring groups records by reported match count. The figure does not include a non-synergistic baseline and does not demonstrate a synthetic-lethal mechanism in the tested combinations.

B. Cluster-Aware OPC/RPC Comparisons

Overall mean OPC was 54.57 versus 22.26, a difference of 32.31 (95% cell-cluster CI 17.29–47.34; $g = 0.499$; BH $q = 7.54 \times 10^{-5}$). Mean RPC was 87.57 versus 33.42, a difference of 54.15 (CI 29.72–78.58; $g = 0.525$; $q = 4.73 \times 10^{-5}$). Figure 2 reports all original labels and group sizes. Liver has one synergistic record; Pleura has one cell cluster, so their primary intervals are NE. Sparse labels are retained rather than silently removed.

C. Sensitivity to Annotation Scale

Within the pathway-mapped subset, source-only standardized OPC/RPC differences were 16.77/24.52. Adding target/pathway size and degree reduced the differences to 2.08 (bootstrap CI 1.23–2.81) and 2.24 (1.08–3.35). Full common support retained 188,274/195,385 records (96.36%); 472/500 bootstrap draws were valid. Support populations and weights differ across schemes, so attenuation is not a causal decomposition.

Fully standardized Jaccard and density differences were 0.00583 (CI -0.00164 to 0.01366) and 6.19×10^{-5} (-2.55×10^{-4} to 4.45×10^{-4}). Both intervals include zero. Restricting the density denominator to Reactome pathways also gave an interval spanning zero. Thus raw-count separation is sensitive to annotation opportunity; it does not establish a scale-independent pathway effect.

D. RPC Threshold Profiles

For RPC $>$ 0, 31,942/159,032 retained records were synergistic (20.09%); for RPC $>$ 500, the proportion was 1,445/2,836 (50.95%). At $>$ 1000, only 160 records remained. Figure 3 shows non-monotonic class proportions and declining support. The dotted reference is the full-cohort synergy proportion (16.98%). These are in-sample descriptive profiles, not classifier validation or a justified optimal cutoff.

IV. Limitations

SL annotations depend on cell context, and drug inhibition is not equivalent to gene knockout. The SL groups have incompletely established selection comparability: uniform annotation of the 36,688 parent-table synergistic records yielded 44.32%, versus 61.37% in the further-selected reported subset. Curation, PubChem coverage and target versions may affect this comparison with the 30.18% background. Uneven annotation, non-equivalent original tissue labels and residual crossed dependence remain limitations. Coarse standardization and one-way clustering do not fully resolve confounding, and no degree-preserving network randomization, independent prediction test or context-specific functional validation is claimed.

V. Conclusions

SynDrugMD describes SL-reference coverage and pathway relationships in combination data. The reported selected SL subset has 61.37% annotation coverage, with a 30.18% descriptive non-synergistic background subject to selection limitations. Corrected statistical comparisons show raw OPC/RPC differences, but scale adjustment substantially attenuates them and normalized adjusted intervals span zero. Count-supported threshold profiles remain descriptive. The evidence supports annotation associations, without establishing causal mechanisms, robust prediction or clinical efficacy.

VI. Data Availability

Curated study data are available from the project repository at xxx. Public input-resource addresses are provided in the Methods and references. Detailed statistical tables and analysis code accompany this revision.

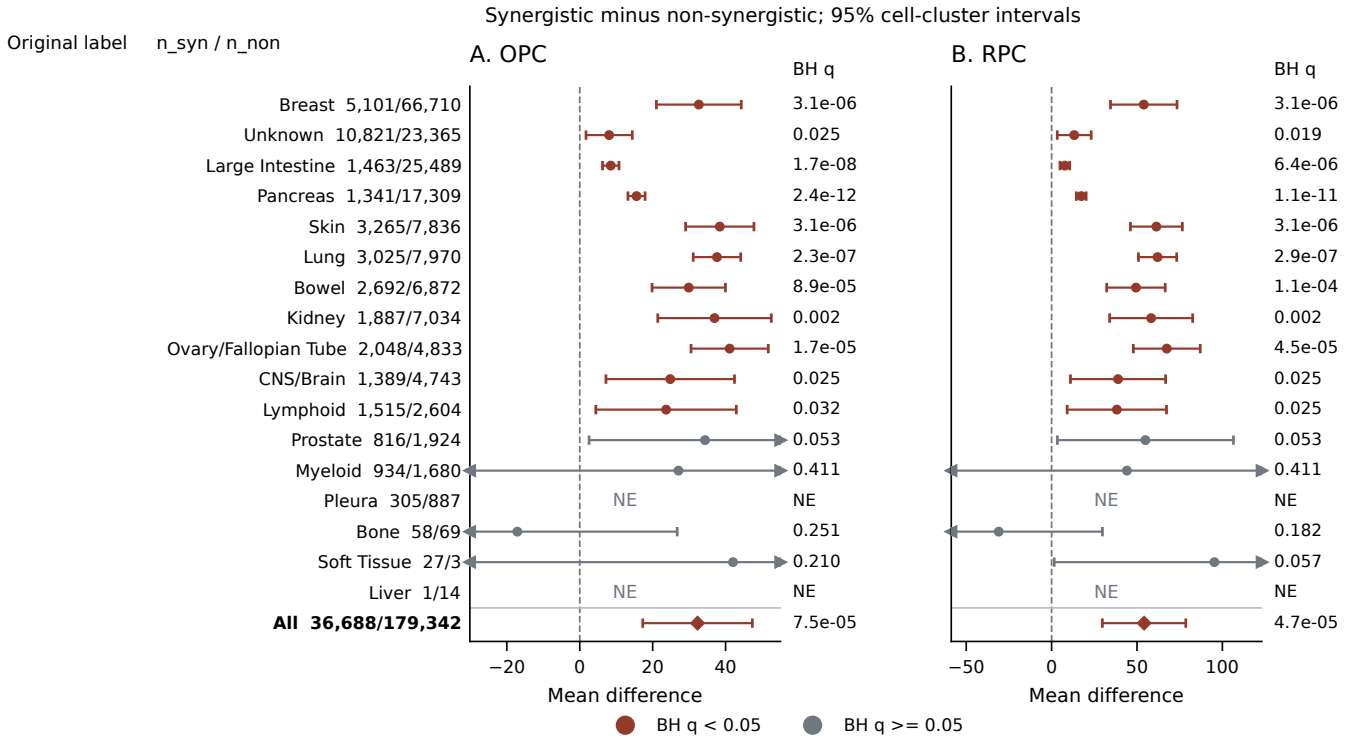


Fig. 2. Mean synergistic-minus-non-synergistic differences in OPC (A) and RPC (B), with 95% cell-cluster intervals and BH q-values. Labels show synergistic/non-synergistic record counts. All 17 original tissue/lineage labels are retained, including Unknown; Bowel and Large Intestine remain distinct. NE denotes insufficient records or cell clusters, not a negative result. Arrows indicate intervals extending beyond the displayed axis range; complete values are in the statistical tables. BH families include the overall comparison and all labels, with 32 estimable tests out of 36 planned per family.

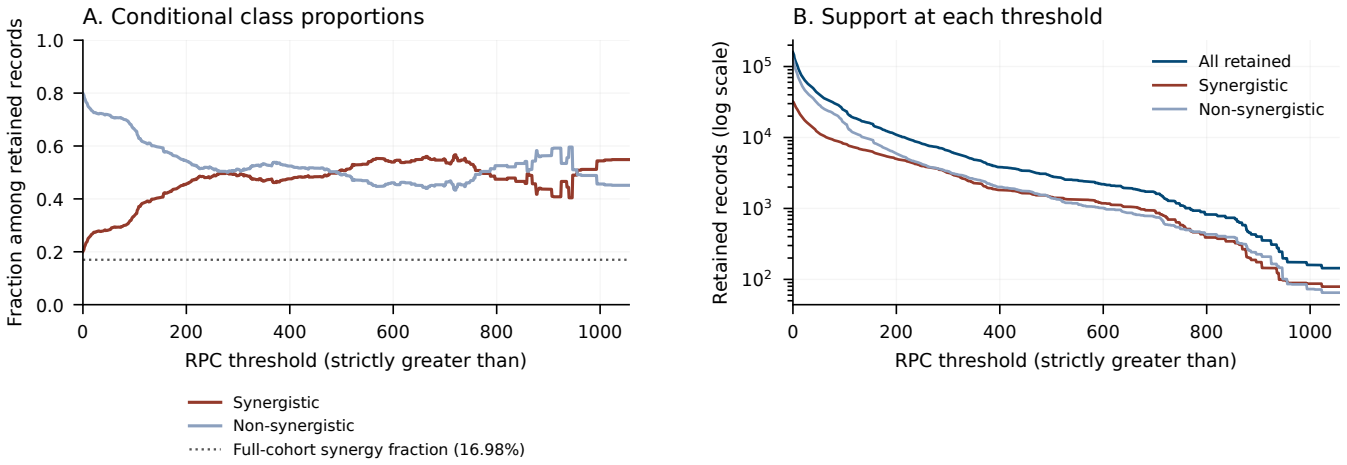


Fig. 3. RPC thresholds use the same denominator in the formula and plots. (A) Class proportions among records with RPC strictly greater than each threshold; the dotted line is the unfiltered 16.98% synergy proportion. (B) Retained total and group counts on a log scale. Display is limited to thresholds with at least 100 retained records; all thresholds are tabulated separately. No 0.5 chance baseline or independent predictive interpretation is used.

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